Supplementary Materials

Mathematical Proofs for the

Geometric-Functional Consistency Framework

Yuhan Zhang

Department of Chemical Engineering and Pharmacy,
Guangling College, Yangzhou University

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Notation and Definitions

Let X = {x_1, ..., x_N} be an embedding of N proteins in R^d.
Let D be the pairwise Euclidean distance matrix, D_{ij} = ||x_i - x_j||.

Spatial graph G(r) = (V, E(r)):
V = {1, ..., N},   E(r) = {(i,j) : 0 < D_{ij} < r}

Community partition C(r) = {C_1, ..., C_K} via greedy modularity optimisation on G(r).

T_v = set of GO terms annotated to protein v.

Functional purity of a community C:

$$\mathrm{purity}(C) = \frac{\max_t |\{v \in C : t \in T_v\}|}{\sum_t |\{v \in C : t \in T_v\}|}$$

Mean functional purity at threshold r:

$$P(r) = \frac{1}{|C(r)|} \sum_{C \in C(r)} \mathrm{purity}(C)$$

G-F Score over integration interval [r_min, r_max]:

$$\mathrm{GF}(X) = \frac{1}{r_{\max} - r_{\min}} \int_{r_{\min}}^{r_{\max}} P(r)\, dr$$

Rescaling operator:

$$R_\sigma(X) = \frac{\sigma}{\mathrm{std}(X)} \cdot X, \ \text{ where } \ \mathrm{std}(X) = \sqrt{\frac{1}{Nd} \sum_{i,k} X_{ik}^2}$$

Proposition 1

(Well-definedness and bounds of G-F purity)

For any embedding X and any community C containing at least one annotated protein:

purity(C) in [0, 1]

Consequently, P(r) in [0,1] for all r > 0, and GF(X) in [0,1] for any integration interval with r_max > r_min.

Proof.

Let n_t(C) = |{v in C : t in T_v}| be the count of proteins in C annotated with GO term t. The set of active terms is T(C) = {t : n_t(C) > 0}. By definition:

$$\mathrm{purity}(C) = \frac{\max_{t \in T(C)} n_t(C)}{\sum_{t \in T(C)} n_t(C)}$$

Since max_t n_t(C) <= sum_t n_t(C), we have purity(C) <= 1.
Since n_t(C) >= 1 for all active terms, purity(C) > 0.
For unannotated communities, purity(C) = 0 by convention.

P(r) is a mean of values in [0,1], hence P(r) in [0,1].
GF(X) is a normalised integral of P(r) over a positive-length interval; since P(r) in [0,1], GF(X) in [0,1].

QED

Proposition 2

(Rescaling is a uniform distance scaling)

For any non-degenerate embedding X (std(X) > 0) and target sigma > 0, let X' = R_sigma(X). For all i, j:

||x'_i - x'_j|| = (sigma / std(X)) * ||x_i - x_j||

Proof.

X' = alpha * X where alpha = sigma / std(X) > 0. Then:

$$\|{x'}_i - {x'}_j\|^2 = \sum_k (\alpha X_{ik} - \alpha X_{jk})^2 = \alpha^2 \|x_i - x_j\|^2$$

Taking square roots (alpha > 0):

||x'_i - x'_j|| = alpha * ||x_i - x_j||

All pairwise distances are multiplied by the same positive scalar alpha. This is a uniform scaling. QED

Theorem 1

(G-F Score scale invariance under rescaling)

Let X and X' = R_sigma(X). Let alpha = sigma/std(X). Then:

(a) P'(r) = P(r/alpha) for all r > 0

(b) The G-F Score is invariant: for any fixed integration interval [r_min, r_max], the G-F Scores computed on X' and X are identical when both are rescaled to the same target standard deviation.

Proof of (a).

By Proposition 2, D'_{ij} = alpha * D_{ij}. Edge (i,j) exists in G'(r) iff 0 < D'_{ij} < r, equivalent to 0 < D_{ij} < r/alpha. This is the edge condition for G(r/alpha). Therefore:

G'(r) = G(r/alpha)   (same graph topology)

Community detection is deterministic and depends only on graph topology, so C'(r) = C(r/alpha). Purity depends only on community membership and GO annotations, hence P'(r) = P(r/alpha).

Proof of (b).

In the pipeline, all methods are rescaled to TARGET_STD = 0.3 before G-F curve computation. After rescaling, any two methods with the same coordinate shape but different original scales produce identical rescaled embeddings, hence identical G-F curves and G-F Scores. QED

Theorem 2

(Subinterval decomposition of G-F Score)

Let a < c < b. The G-F Score over [a,b] decomposes as:

$$\mathrm{GF}([a,b]) = w_1 \cdot \mathrm{GF}([a,c]) + w_2 \cdot \mathrm{GF}([c,b])$$

where w_1 = (c-a)/(b-a), w_2 = (b-c)/(b-a), with w_1+w_2=1.

Corollary:

min(GF([a,c]), GF([c,b])) <= GF([a,b]) <= max(GF([a,c]), GF([c,b]))

Proof.

By definition, splitting the integral at c:

$$\int_a^b P(r)\,dr = \int_a^c P(r)\,dr + \int_c^b P(r)\,dr$$

Substituting the normalised integrals:

$$\mathrm{GF}([a,b]) = \tfrac{c-a}{b-a}\,\mathrm{GF}([a,c]) + \tfrac{b-c}{b-a}\,\mathrm{GF}([c,b])$$

This is a convex combination (w_1, w_2 > 0, w_1+w_2=1).
The corollary follows from the standard property that a
convex combination lies in the closed interval of its terms.

QED

Theorem 3

(Plateau width and G-F Score stability)

Let method m have plateau width W_m (relative threshold theta = 0.80) and peak purity p*_m. If [r_min, r_max] is fully contained within the plateau region, then:

$$\boxed{\mathrm{GF}_m \geq \theta \cdot p^*_m}$$

Furthermore, for any two methods m_1, m_2 with plateau regions containing [r_min, r_max]:

$$\boxed{|\mathrm{GF}_{m_1} - \mathrm{GF}_{m_2}| \leq |p^*_{m_1} - p^*_{m_2}|}$$

Proof.

Within the plateau, P(r) >= theta * p*_m for all r. If [r_min, r_max] is a subset of this region:

$$\boxed{\mathrm{GF}_m = \frac{1}{r_{\max} - r_{\min}} \int_{r_{\min}}^{r_{\max}} P(r)\, dr \geq \frac{1}{r_{\max} - r_{\min}} \int_{r_{\min}}^{r_{\max}} \theta\, p^*_m\, dr = \theta\, p^*_m}$$

For the second part, within both plateaus the purity difference is bounded by the peak purity difference, and integration preserves this bound.

Practical implication: methods with wider plateaus have G-F Scores more robust to the choice of integration interval.

QED

Theorem 4

(Random baseline convergence)

Under a random permutation pi of node labels, the expected G-F purity at threshold r converges to the community-size-weighted mean of the marginal GO term distribution:

b{E}_\pi[P_\pi(r)] \;\to\;\sum_v \frac{|C_v(r)|}{N}\; p_{\mathrm{unif}}$\quad\text{as } N

where p_unif = max_t f_t / sum_t f_t is the uniform purity from global GO term frequencies f_t.

Furthermore, Var[P_pi(r)] = O(1/N), so the random baseline concentrates around its expectation.

Proof.

Under permutation pi, community C_v(r) depends only on embedding geometry (unchanged). The GO terms assigned to C_v(r) form a simple random sample of size |C_v(r)| from the global annotation multiset.

For term t with global frequency f_t:

$$\mathbb{E}[n_t(C_v)] = f_t \cdot |C_v(r)|/N$$

By multivariate hypergeometric concentration (Chvatal 1979), n_t(C_v) concentrates around its mean for large |C_v|. Therefore purity(C_v) -> p_unif as |C_v| -> infinity.

The mean purity P_pi(r) is an average of K such terms, each concentrating around p_unif. By variance bounds for sampling without replacement: Var[P_pi(r)] = O(1/N).

The G-F Score under permutation: E[GF_pi] -> p_unif with Var[GF_pi] = O(1/N) by dominated convergence.

Practical implication: the randomisation baseline (10 shuffles) estimates p_unif. Methods with GF significantly above this baseline capture genuine geometric-functional alignment. QED